

ARISTA



CloudVision Workshop 2026

Section 2 : Onboarding with Essential Studios

Device Onboarding

Onboarding Methods

Zero Touch Provisioning (ZTP)

ZTP enables network devices to achieve full operational readiness through automatic discovery, image installation, and configuration without manual intervention.

Requirements:

- Management Connectivity
- DHCP
- DHCP Option 67 (CVP On Prem)
- DNS
- Reachability to CloudVision

Manual Onboarding

There are many use cases where ZTP may either not be possible or not part of your operational workflow and you need to onboard a device manually. You can initiate manual onboarding via the device CLI.

Requirements:

- Reachability to CloudVision
- DNS
- Console Cable and A Dream

Manual Onboarding

Console

Device
Configuration

Device
Registration

Download
Config

Production
Ready ✓

Onboarding

Generate and run a customized script to onboard a device to CloudVision.

Select a validity period for the onboarding token, optionally select a [configured profile](#), and click Generate. Run the script to verify Streaming Agent installation, synchronize the device clock, pass the onboarding token to the appropriate directory, and onboard the device.

If the management interface of the device needs to reach CloudVision through a VRF, make sure to configure and select an onboarding profile with the appropriate VRF name.

Token will expire after 1 day



Select an Onboarding Profile



Regenerate



bash

```
(export AUTH_TOKEN="eyJhbGciOi..." TOKEN_PATH="/tmp/cv-onboarding-token"; echo "$AUTH_TOKEN" > "$TOKEN_PATH" && curl --fail -H "Authorization: Bearer $AUTH_TOKEN" https://www.cv-prod-us-4.arista.io/scripts/onboard.sh | bash -s -- --smashexcludes=ale,flexCounter,hardware,kni,pulse,strata -cvaddr=apiserver.arista.io:443 -cvauth=token-secure,"$TOKEN_PATH" -taillogs); exit
```

Paste the script into the EOS CLI and press Enter. The device will onboard to the [Inventory and Topology Studio](#) as a network update that you'll need to accept.

[Copy Token](#)

[Copy Script](#)

Zero Touch Provisioning **Onboarding**

ZTP Process



- Zero human interaction required after racking the switch
- Consistent, repeatable deployments across all sites
- Ship switches directly — no staging or pre-configuration
- CloudVision integration for lifecycle management from day one



Studios Overview

Core Fundamentals of Automation

1. Inventory

A subset of devices upon which we will be operating or creating configuration. Includes authentication and potentially some topological information.

2. Data Model

A Data model is a constrained set of questions to be asked of the user which will provide the needed data for configuration generation. It is well defined within the automation framework component.

3. Data

The answers to the data model questions for each device. This data may, depending on the data model, specify individual values for specific devices, or may answer global questions that apply to all devices

4. Code

All automation frameworks will contain some form of code and/or templates that convert data into a proper configuration snippet. This may be a complete config, or a subset of a full config depending on the framework.

“Studios” Introduction

What is Studios?

Studios is a combined framework that generates configuration configlets for sections of the configuration.

The use of multiple studios provides a complete configuration. Studios generate both dynamic and static configuration for devices configured in the **inventory**. The combination of these configuration snippets comprises the complete device configuration.

- Fabric provisioning, (underlay and overlay)
- Date/Time related configuration
- Interface definitions
- Authentication
- Statically defined configuration

For unique cases, custom studios can be defined.

Why use Studios?

Static device configuration can be cumbersome and not easily abstracted due to unique requirements across devices.

Device configurations also include static values unique across all devices, like a hostname or management interface IP addresses or management ACLs

Studios provides mechanisms to define global values inside dynamic device specific configuration blocks, and static unique configurations, combined together to form a full configuration.

Workspaces Overview

Workspaces

What are Workspaces?

An isolated staging environment where you build and review proposed network configuration changes — without affecting production.



Sandbox

Like a dev sandbox — experiment freely



Config Session

Like a config session in EOS / NOS CLIs



Git Branch

Branch, compare, and merge — just like a pull request

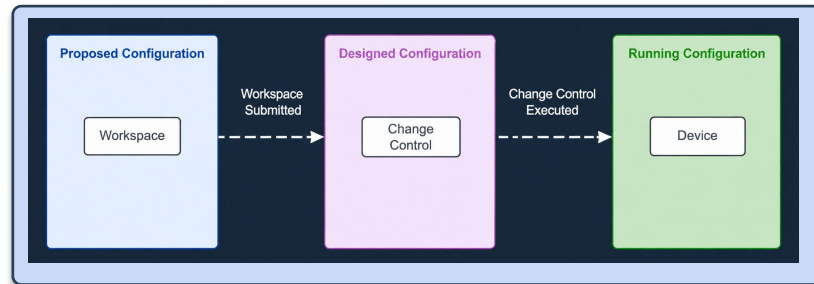


Save & Commit

Batched operations, like some friendly neighborhood Firewall GUIs

How does it work?

Once a workspace is successfully validated and submitted, its changes become part of CloudVision's designed configuration and is passed to Change Control for approval and deployment.



Essential Studios - Inventory & Topology

What is it?

A foundational studio that maps back to the core fundamentals of automation—establishing a reliable, structured device inventory.

What does it do?

Adds devices directly into the active management INVENTORY.

Builds and dynamically updates the physical topology using LLDP neighbor details.

When to use it?

Executed as the initial phase of the onboarding workflow to register hardware and map link connectivity across the network fabric.

CloudVision Walkthrough



CloudVision Lab - 01



Essential Studios

CloudVision Tags - Label:Value Pairs



What is it?

A tag is a `Label:Value` pair used for grouping devices or interfaces under a common characteristic.

There are three types of tags: custom tags, CloudVision tags, and external tags.

What does it do?

Tags let CloudVision operate on intent-based groups rather than hard-coded lists of switch names.

Drive CloudVision Studios.

Represent network hierarchy.

Useful for search and filtering.

Drive dashboards, events and notifications.

When to use it?

CloudVision Tags are automatically generated and assigned to devices and interfaces.

Use custom tags when CloudVision's built-in/system tags don't express a piece of metadata that is meaningful to your network operations or automation.

Essential Studios - Static Configuration

What is it?

A flexible bridge between low-level CLI commands and modern tag-based abstraction.

It is used to manage static configuration for devices, provide configuration not created by other studios, and reconcile designed vs. running configurations.

What does it do?

Applies "static" configuration directly to target devices.

Serves as a fallback mechanism to inject raw CLI commands safely.

When to use it?

Deploying features where standard, built-in system studios do not fit the specific requirements.

Utilized in custom deployment scenarios.

Essential Studios - Software Management

What is it?

The Software Management Studio is used to manage and upgrade EOS device images and other software, including the Streaming Agent.

What does it do?

Handles version control by assigning image versions to devices based on tags.

When to use it?

Before switches are on-boarded into CloudVision to set EOS images to devices in ZTP mode.

To upgrade EOS image on provisioned* devices.

Change Control

Change Control Overview

Change Control is used to manage updates and push configuration changes to your network. Use Change Control to approve and execute changes made in Workspaces and to run actions either independently or alongside configuration changes.

CHANGE CONTROL PROCESS



KEY CAPABILITIES

Stages & Execution

Group actions into stages.
Execute in series or parallel.
Nest sub-stages for complex orchestration.

Approval Controls

Non-Author Review enforces separation of duties. Approve and Execute toggle for streamlined workflows.

Rollback & Scheduling

Stop execution mid-flight.
Rollback revertible actions (Set Config, Set Image).
Schedule execution time.

CloudVision Lab - 02



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Thank You

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